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Predicting the Sustainability of Pickleball Competitions as a New Sport from the Behavioral Intention of Pickleball Players

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Abstract: This study focused on predicting the behavioral intentions of pickleball players. It analyzed the predictability of pickleball players' behavioral beliefs on their attitudes, normative beliefs on subjective norms, control beliefs on perceived behavioral control, attitudes on behavioral intentions, subjective norms on behavioral intentions, and perceived behavioral control on behavioral intentions. The subjects were pickleball players, and purposive sampling was adopted for the questionnaire survey. A total of 226 valid questionnaires were collected. The data were analyzed with descriptive statistics and structural equation modeling. The results show that behavioral beliefs had a significant impact on attitudes; normative beliefs had a significant impact on subjective norms; control beliefs had a significant impact on perceived behavioral control; attitudes had no significant impact on behavioral intentions; subjective norms had a significant impact on behavioral intentions; and perceived behavioral control had a significant impact on behavioral intentions. In the future, researchers are suggested to extensively collect empirical data from players of different levels in pickleball competitions through interviews or on-site observations, and the analyses based on relevant theories can be conducive to the predictive analysis of the behavioral intentions of pickleball players, and practical suggestions are proposed regarding the sustainability of pickleball competitions.

Keywords: pickleball; sustainability; behavioral intentions; attitudes; subjective norms

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1. Introduction

According to the Sports Status Survey Results 2018 of the Sports Administration, the proportion of the population participating in sports in Taiwan was 83.1%, and in particular, the proportion of the population who regularly exercised reached a record high of 33.5%. It also showed that the female sports culture was mature, and the elderly sports population was far higher than the average of the entire population, which indicated that sports are becoming more and more prosperous in Taiwan [1]. It can be seen that ball-based sports or games not only are popular with the people but can also increase the willingness to participate in indoor ball games that are not constrained by weather factors. Almond [2] divided ball games into four categories according to their attributes, including target sports, net/wall sports, fielding/run-scoring sports, and invasive sports. Target sports have the characteristics of aiming and are intended to let the ball enter a certain range via the equipment; in net/wall sports, nets or walls are used as the medium to distinguish the progress of the game; in fielding/run-scoring sports, one side attacks and the other side defends alternately, and when the attacking side seeks to obtain points, the defending side prevents it from scoring; and invasive sports involve close combat between the two sides.

This classification enables a further understanding of the diverse characteristics of ball games [2].

In recent years, with the prosperity of online media, people have gradually learned of all kinds of new ball games through newspapers and media. Pickleball, a ball game that is still unfamiliar to the Taiwanese people, has been slowly promoted in Taiwan in the era of the rise of the Internet. Pickleball originated in Seattle, United States, in 1965 [3]. By referring to the game rules of tennis, billiards, and badminton, the inventor Joel Pritchard et al. started this new sport with family, neighbors, and friends on badminton courts in their own backyards. Pickleball started as an unintentional activity, and its process is both casual and fun [4]. At present, pickleball in Taiwan is still mainly promoted by the Chinese Taipei Pickleball Association, the Pickleball Committee of the Chinese Taipei University Sports Federation, and the Pickleball Associations of various counties and cities, while some colleges and universities have also incorporated pickleball into their sports curriculum. In addition, the related promotional units of pickleball hold different events so that pickleball lovers can exchange their skills and pursue their own goals. The 2019 International Pickleball NPRP Tournament in Taichung attracted global pickleball players to participate in the competition, and this tournament became a media favorite to promote sports tourism. In other words, the contemporary combination of sports and tourism gives players from Taiwan and abroad stronger motivation to participate in such events.

People often engage in behavior based on a certain intention, which is the determinant factor in determining whether the behavior occurs. The theory of planned behavior in social psychology focuses on people's intention to engage in a behavior; that is, people will first consider the results of the behavior before deciding whether to get involved in it [5]. As can be seen from the relevant literature, the research topics related to pickleball mostly focus on the issue of elderly participation in pickleball ([4–11]). For example, in a study of older persons by [9], the results support two possible conclusions: (1) that motivation and benefit are different and potentially disconnected constructs and (2) that the benefit of competition among this sample of older persons is understood through the lens of personal mastery, whereby the demonstration of that mastery is only possible through competition.

The results showed that respondents at the highest psychological continuum model levels had played pickleball for at least one year and played at least 10 times per month. Although fitness and socialization were reported as the most important motives for the entire sample, competition and skill mastery were rated significantly higher for respondents at higher psychological continuum model levels. Wray, Ward, Nelson, Sulzer, Dakin, Thompson, Vierimaa, Das Gupta, and Bolton [11] conducted a feasibility study to evaluate the impact of a six-week pickleball intervention on the measures of muscle function, cognitive function, perceived pain, and cardio-metabolic risk, as well as several psychosocial factors that contribute to adherence in sedentary rural participants. The results showed that participants improved their vertical jump and cognitive performance, and they reported a decrease in self-reported pain, which suggested improved physical and cognitive health across the sample. Participants also reported high levels of satisfaction and demonstrated good adherence over the duration of the study.

Ryu, Heo, Lee, Kim, and Kim [12] examined the association between authenticity and psychological functioning in older adults playing pickleball. Forrester [13] analyzed the injuries caused by pickleball sports in the emergency room from 2013 to 2017 and found that patients 50 years old or older accounted for 90.9% of the cases, and 50.4% were males.

It can be seen from the abovementioned literature that relevant research on the participation of the elderly in pickleball was mostly explored from the perspectives of their participation motivation, leisure involvement, and physical and psychological dimensions. However, the focus of attention of pickleball players is the ease or difficulty of completing the game. In other words, the non-motivational factors of skills, abilities, and resources beyond the control of the individual are worth exploring. The theory of planned behavior provides a theoretical basis to explore the attitude of pickleball players in choosing this sport, as well as the factors that affect the choice. Biddle and Nigg [14] also pointed out

that the theory of planned behavior is a widely used and effective theoretical framework for understanding and predicting leisure sports behaviors. It is also known from related research that the theory of planned behavior is extensively used to explore sports behavior issues; for example, Pao [15] tested the behavioral intention pattern of Taitung paragliding tourists, and the results showed that the "attitudes", "subjective norms", and "perceived behavioral control" of Taitung paragliding tourists all had a positive and significant impact on "behavioral intentions", and subjective norms had a mediating effect on behavioral intentions through attitudes. The feasibility of using the theory of planned behavior for basic theoretical exploration and dimension verification in sports-related fields can be seen from the above related studies.

For a long time, the theory of planned behavior has provided a good theoretical basis for research on human behaviors. Bosnjak, Ajzen, and Schmidt [16] pointed out that as of April 2020, the theory of planned behavior has been subject to empirical scrutiny in more than 4200 papers, as referenced in the Web of Science bibliographic database, rendering it one of the most applied theories in social and behavioral sciences. Relevant research reveals that the theory of planned behavior has received broad attention in various areas, such as health sciences, environmental science, business and management, and educational research. The theory of planned behavior is widely used in different research fields, such as illustrated by the study by Ulker-Demirel and Ciftci [17], which pointed out that the theory of planned behavior has widely been extended to different contexts in the framework of tourism, leisure, and hospitality management, which are different from most social psychology theories in terms of feasibility. The theory of planned behavior posits three determinants of intention, which indirectly apply their effects on behavior with an effect on intention. The first is the attitude toward a behavior, which refers to the degree of evaluation or appraisal of a person from favorable to unfavorable in terms of the behavior in question. The second predictor is a social factor known as the subjective norm, which refers to perceived social pressure to perform or not perform a behavior. The third determinant of intention is the measure of perceived behavioral control, which refers to the perceived ease or difficulty of performing the behavior and reflects past experiences as well as anticipated impediments and obstacles [17]. The factors affecting behavioral intentions can be known from the abovementioned definitions, which are taken as the theoretical basis for the hypotheses in this research.

As pickleball becomes more popular in Taiwan, exploration of the behavioral intentions of participants, as well as the affecting factors, will help further promote pickleball events in Taiwan. Furthermore, under the concept of sustainability, as the number of people who know about pickleball increases, the more helpful it will be to build a pickleball event environment with less impact on the environment and society. Relevant literature indicates the feasibility of applying the theory of planned behavior in the sports field.

Yeh [18] held a similar view and argued that the theory of planned behavior has long been applied to related research abroad, while the use of this theory only began in Taiwan roughly after 1995. Furthermore, it has been widely applied in research on consumer behaviors, scientific and technological information, medical health, knowledge management, and career choice. According to the literature review, this study found that in recent years, the content of the theory of planned behavior to explore the field of leisure sports and tourism has gradually been enriched, for example, in the field of leisure tourism.

For example, in the field of leisure tourism, ref. [19] analyzed the behavioral tendencies of Taiwanese tourists in Kinmen, and the results showed that tourists' attitudes, subjective norms, and perceived behavioral control had a positive and significant effect on their behavioral intentions. Chiang [20] took tourists to the Bunun Leisure Farm as the research subjects to explore the tourism relationship between the destination attractiveness of the aboriginal sightseeing places, place attachment, and behavioral intention.

In terms of applications in sports-related fields, ref. [21] used the theory of planned behavior to understand the behavioral intentions of junior high school students engaging in leisure sports, and the results showed that the subjective norms of junior high school

students engaging in leisure sports had a direct and positive impact on their behavioral intentions. Moreover, the addition of perceived value had a significant positive impact on the overall theory of planned behavior, meaning that it positively regulated the constructs of behavioral attitudes, subjective norms, and perceived behavioral control. Furthermore, it directly and positively affected behavioral intentions and effectively improved the willingness of junior high school students to engage in leisure sports. Sung [22] aimed to understand the behavioral intentions and behaviors of handball players from the perspective of the theory of planned behavior, and the results showed that behavioral attitudes and subjective norms had a high positive correlation, subjective norms and perceived behaviors had a moderate positive correlation, and behavior attitudes and perceived behavioral control had a moderate positive correlation. Chang [23] explored the behavior of college students participating in surfing according to the theory of planned behavior, and the results showed that the combination of behaviors and beliefs could effectively predict attitudes, the combination of norms and beliefs could predict subjective norms, and the combination of controls and beliefs could effectively predict perceived behavioral control. Based on the theory of planned behavior, Yen, Hsu, and Pan [24] constructed a mediation model to investigate the behavioral intention of tourists in water sports in the Kenting area by taking the modified attitude of related theories as a mediating variable, and their research results echoed the views of [25], which pointed out the effectiveness of the theory of planned behavior in predicting sports behavioral intentions through empirical evidence. The empirical research findings of [26] also showed that the theory of planned behavior had significant predictive power in establishing the behavioral intention of participating in exercise habits. The behavioral intention of participating in sports can be predicted by three variables: attitude, subjective norms, and perceived behavioral control. The variable of perceived behavioral control is commonly used, as it has the highest influence on the behavioral intention to engage in regular exercise. Kouthouris and Spontis [27] also pointed out the feasibility of applying the theory of planned behavior to research related to outdoor sports.

Regarding the relationship between the various dimensions of the theory of planned behavior, it is pointed out that behavioral beliefs have a significant impact on attitudes; for example, ref. [28] pointed out that behavioral beliefs had a significant impact on attitudes. Furthermore, normative beliefs have a significant impact on subjective norms; for example, ref. [29] used the theory of planned behavior to explore the health literacy of construction workers, and the results showed that normative beliefs and compliance motivation had a significant positive impact on subjective norms. Ajzen [17] found that control beliefs had a significant impact on perceived behavioral control and pointed out that the measure of perceived behavioral control could be composed of the product of control belief and control force. In addition, related research has also indicated that attitudes, subjective norms, and perceived behavioral control had a significant impact on behavioral intentions; for example, ref. [30] investigated the effects of attitudes toward fashion on the participation of adolescent students in physical activity and sport and found that there were significant relationships between the components of the theory of planned behavior (e.g., attitude, subjective norms, and perceived behavioral control) regarding fashion with intention to engage in physical activity and actual physical activity behavior in adolescents. A study [31] focused on taekwondo, where the main purpose was to examine the structural relationships among mentoring, attitudes, subjective norms, perceived behavioral control, and career pursuit intentions by applying the theory of planned behavior. In addition, the study investigated the moderating influence of taekwondo identification on these relationships. There were positive impacts of attitude on career pursuit intention, subjective norms on career pursuit intention, and perceived behavioral control on career pursuit intentions.

The above related literature shows that the relationship between variables has been confirmed by previous studies. In addition, since pickleball is a new sport promoted in Taiwan, it is meaningful to use the theory of planned behavior to conduct related discussions and empirical analysis. Therefore, this study used the theory of planned behavior to explore the behavioral models of participants in the 2019 International Pickleball NPRP Tournament.

2. Research Method

2.1. Research Structure

Based on the literature review, research purpose, and research structure, the following hypotheses are proposed by this study, and the research structure is shown in Figure 1.

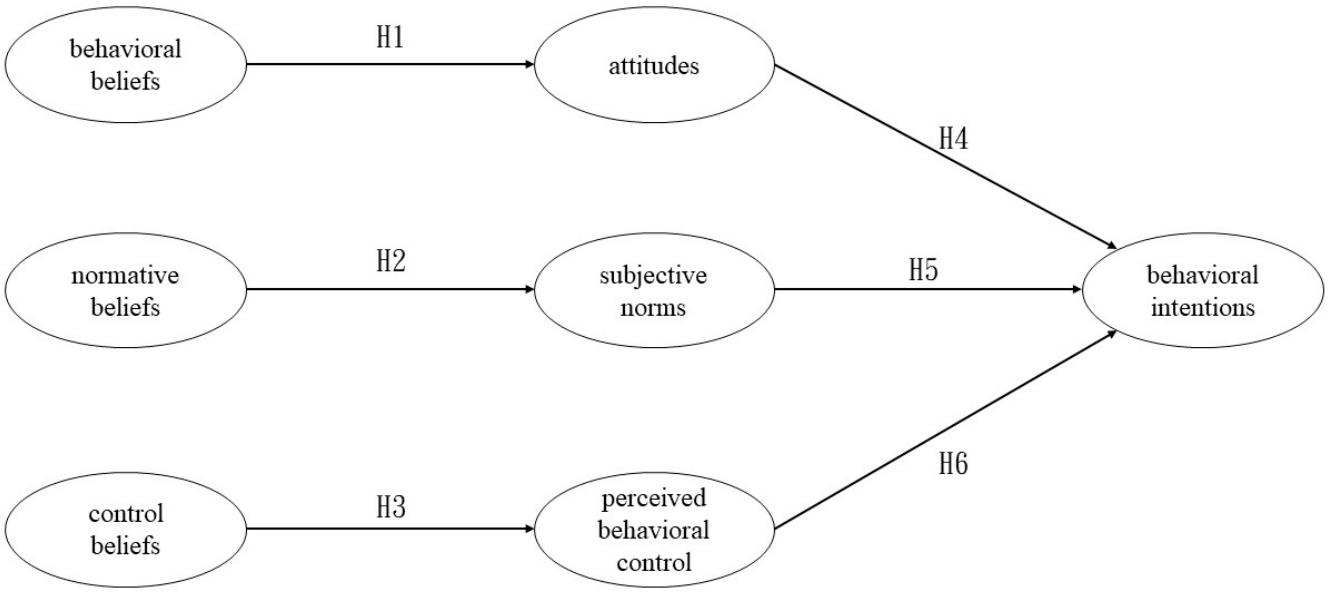

Figure 1. Research structure.

2.2. Research Hypotheses

H1: The behavioral beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on attitudes. H2: The normative beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on subjective norms. H3: The control beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on perceived behavioral control. H4: The attitudes of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on their behavioral intentions to participate. H5: The subjective norms of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on their behavioral intentions to participate. H6: The perceived behavioral control of players in the 2019 International Pickleball NPRP Tournament has a significant positive impact on their behavioral intentions to participate.

2.3. Research Questions

The research aims were (1) to explore whether the behavioral beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant impact on attitudes; (2) to explore whether the normative beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on subjective norms; (3) to explore whether the control beliefs of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on perceived behavioral control; (4) to explore whether the attitudes of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on their behavioral intentions to participate; (5) to explore whether

the subjective norms of players in the 2019 International Pickleball NPRP Tournament have a significant positive impact on their behavioral intentions to participate; and (6) to explore whether the perceived behavioral control of players in the 2019 International Pickleball NPRP Tournament has a significant positive impact on their behavioral intentions to participate.

2.4. Research Subjects

This study took the players in the 2019 International Pickleball NPRP Tournament as the research subjects, and the intentional sampling method was adopted for the distribution of questionnaires to 260 players from 7 countries in the gymnasium of National Chung Hsing University, Taichung, from 21 to 22 December 2019. The questionnaires of this study have Chinese, English, and Japanese versions. In order to ensure the clear semantics of the questionnaires, Tomo Ikechan, a Japanese doctoral student from the School of Operation and Management of Asia University, was specifically requested to assist in translating the questionnaires. The distribution of the questionnaires was approved by the tournament organizer beforehand. Before the questionnaires were distributed to the participating teams from various countries, the purpose of the questionnaires was explained to the players who filled the questionnaires, indicating that this study specifically designed the questionnaires for this international competition for research and analysis purposes, with no other intentions. A total of 255 questionnaires were recovered, and the recovery rate of the questionnaires was 98%. The returned questionnaires were first checked to see if there were invalid questionnaires with blank, incomplete, or single answers. There were 226 valid questionnaires, and the rate of valid questionnaires was 88.6%.

2.5. Research Tools

The content of the questionnaire in this study was mainly compiled with reference to the relevant literature [32] and the modification of the questionnaire. The questionnaire was divided into 2 parts, with a total of 45 items, including 8 items on basic personal information, 7 items on behavioral beliefs, 5 items on attitudes, 6 items on normative beliefs, 4 items on subjective norms, 7 items on control beliefs, 4 items on behavioral intentions, and 4 items on perceived behavioral control. This study used a 7-point Likert scale, and the options for each item were given a score from 1 to 7 for "Strongly disagree" to "Strongly agree", respectively.

2.6. Data Processing and Analysis

In this study, after the collected questionnaires were analyzed for valid questionnaires and the invalid questionnaires were excluded, the data were archived with Statistical Product and Service Solutions (SPSS) 24.0 statistical software. The distribution of each item was understood through various methods, such as frequency distribution, percentage, Cronbach's α, and median, and then the correlation between variables was analyzed with AMOS 24.0 statistical software.

3. Research Results

3.1. Sample Characteristics

In this study, there were 226 valid samples, and the sample characteristics are shown in Table 1.

Table 1. Sample Characteristics.

Background Variable	Classification Criteria	Number of Samples	Percentage %	Accumulated Percentage %
Gender	Male	141	62.4	62.4
	Female	85	37.6	100.0
Age	20 years old (including) and below	42	18.6	18.6
	21–30 years old	76	33.6	52.2
	31–40 years old	37	16.4	68.6
	41 years old and above	71	31.4	100.0
Nationality	Taiwan	125	55.3	35.0
	Japan	30	13.3	53.5
	Hong Kong	44	19.5	84.5
	Singapore	13	5.8	93.4
	South Korea	6	2.7	96.0
	India	2	0.9	96.9
	United States	4	1.8	99.1
	Others	2	0.9	100.0
Frequency of playing pickleball	1–2 times per month	55	24.3	24.3
	3–4 times	46	20.4	44.7
	5 times or more	125	55.3	100.0
Will you continue to play pickleball in the future?	Yes	199	88.1	88.1
	No	27	11.9	100.0
Educational level	Elementary school	8	3.5	3.5
	Junior high school	7	3.1	6.6
	Senior high and vocational school	38	16.8	23.5
	Junior college	11	4.9	28.3
	College and university	133	58.8	87.2
	Graduate school (including) and above	29	12.8	100.0
Participation level	2.0 Classification	84	37.2	37.2
	3.0 Classification	50	22.1	59.3
	Open tournament	24	10.6	69.9
	Team competition	30	13.3	83.2
	Audience	38	16.8	100.0
Racket price	Below NTD 1000	37	16.4	16.4
	NTD 1001–2000	42	18.6	35.0
	NTD 2001–3000	72	31.9	66.8
	NTD 3001–4000	38	16.8	83.6
	NTD 4001–5000	30	13.3	96.9
	Above NTD 5001	7	3.1	100.0

3.2. Analysis of the Current Status of Theory of Planned Behavior Questionnaire

The analysis results of the questionnaire on the theory of planned behavior are shown in Table 2. There are seven sub-dimensions for the theory of planned behavior in this study, namely, behavioral beliefs, attitudes, normative beliefs, subjective norms, control beliefs, behavioral intentions, and perceived behavioral control. Among the seven sub-dimensions, the sub-dimension of attitudes had the highest average total score of 5.56, followed by behavioral intentions at 5.49, behavioral beliefs at 5.38, subjective norms at 5.29, normative beliefs at 5.22, control beliefs at 5.07, and perceived behavioral control at 4.68, which was the lowest, as shown in Table 2.

Table 2. Analysis of the Current Status of Theory of Planned Behavior Questionnaire.

Item No.	Dimension	Item	Median	Standard Deviation	Bias	Kurtosis	Dimension Average
A1	Behavioral beliefs	I can get knowledge of pickle balls	6.00	1.17	−0.88	1.61	5.38
A2		I can get pickleball skills	5.00	1.23	−0.66	0.60	
A3		I can relax and relieve stress	5.00	1.43	−0.62	−0.23	
A4		I can be an opportunity to connect with family and friends	5.50	1.37	−0.75	0.40	
A5		I can raise my feelings about pickleball	6.00	1.21	−1.06	2.17	
A6		I can support my favorite team	5.00	0.97	−0.76	2.77	
A7		I can support my favorite player	5.00	1.12	−1.05	3.18	
B1	Attitudes	It is good to participate in the pickleball game	6.00	1.08	−0.57	0.98	5.56
B2		I am happy to be in the pickleball game	6.00	1.00	−0.29	−0.13	
B3		Participating in a pickleball game is fascinating	6.00	1.11	−0.50	0.39	
B4		Participating in a pickleball game is fun	6.00	1.03	−0.53	0.90	
B5		Participating in a pickleball game is meaningful	6.00	1.10	−0.63	0.85	
C1	Normative beliefs	A teacher thinks it is good to participate in the ice ax ball	5.00	1.08	−0.07	−0.63	5.22
C2		Parents think it's good to participate in the ice ax ball	5.00	1.14	−0.34	0.43	
C3		Brothers and sisters think it's good to participate in the ice ax balls	5.00	1.22	−0.23	−0.48	
C4		School friends (classmates) think it is a good idea to participate in the ice ax balls	5.00	1.30	−0.34	−0.59	
C5		My friends think it's good to participate in the ice ax ball	5.00	0.82	0.29	−0.32	
C6		Supporters think it is a good thing to participate	5.00	0.97	−0.26	1.46	
D1	Subjective norms	I think others should play pickle balls	5.00	1.05	−0.52	0.43	5.29
D2		I think many people should support pickleball participation	5.00	1.13	−0.60	0.42	
D3		Many people want to participate in the pickleball	5.00	1.16	−0.64	0.63	
D4		I think many people should participate in pickleball games	5.00	1.16	−0.40	−0.10	
E1	Control beliefs	Understand pickleball rules	5.00	1.19	−0.22	−0.72	5.07
E2		I have information about pickle balls	5.00	1.37	−0.30	−0.51	
E3		Have time to play pickleball	5.00	1.48	−0.37	−0.38	
E4		Sufficient physical strength to participate in the pickle game	5.00	1.29	−0.24	−0.75	
E5		I have a friend participating in the pickleball	5.00	1.37	−0.84	0.71	
E6		I have equipment for pickling balls (monitors and computers)	5.00	1.42	−0.64	0.03	
E7		Live TV or Internet video broadcast	5.00	1.26	−0.96	1.63	

Table 2. *Cont.*

Item No.	Dimension	Item	Median	Standard Deviation	Bias	Kurtosis	Dimension Average
F1	Behavioral intentions	I want to participate in a pickleball game	6.00	1.14	−0.82	1.73	5.49
F2		Can participate in pickleball games	6.00	0.96	−0.0	2.36	
F3		Looking forward to participating in pickleball games	5.00	1.02	−0.35	0.63	
F4		Need to participate in pickleball	5.00	1.11	−0.77.	2.13	
G1	Perceived behavioral control	I think it is very difficult to participate in a pickleball	4.00	1.69	−0.07	−1.13	4.68
G2		I think it's easy to join a pickleball	5.00	1.15	0.23	−0.78	
G3		Participating in pickle balls is often what you can do	5.00	1.11	0.04	−0.41	
G4		I want to participate in pickleball whenever possible	5.00	1.05	−0.18	−0.44	

Source: Compiled by this study.

3.3. Structural Equational Modeling

After the valid questionnaires were coded in sequence, the SPSS 24.0 statistical software was used to create a file. After the data obtained by the cumulative number and percentage of the item types were presented, the AMOS 24.0 statistical application was used to verify and analyze the SEM fit.

(1) Confirmatory Factor Analysis

Confirmatory factor analysis was used to examine the extent to which the measurement variable constitutes a latent variable and to test whether the causal pattern of the measurement variable and the latent variable is consistent with the observed data [33,34]. There were a total of 37 items for the dimensions of behavioral beliefs, attitudes, normative beliefs, subjective norms, control beliefs, behavioral intentions, and perceptual behavioral control. The result of CFA indicates that although the standardization coefficients of A6, A7, C6, E6, E7, and G1 were not up to the standard of 0.7, they were still within the acceptable range. As the residuals were all positive and significant, it was obvious that there was no violation of the estimation, as shown in Table 3.

(2) Measurement Model Analysis

The reliability and validity of the questionnaire in this study were analyzed by CFA, and the items in the questionnaire were revised with reference to modification indices (MI). The analysis was performed again [35]. If the value of the measurement error correction index was higher than 3.84 and reached a significant level, it meant that the measurement errors between the two items were correlated, and the measurement errors of the two items could be corrected [34]. In this study, the items with the highest value were deleted among those items with a measurement error correction index value higher than 3.84, and items A2, A6, and A7 of behavioral beliefs; items C5 and C6 of normative belief; items B2 and B5 of attitudes; items E1, E6, and E7 of control beliefs; and item G1 of perceived behavioral control were deleted.

Table 3. Violation Estimation Test Form.

	Estimate	S.E.
A1 <— Behavioral beliefs	0.82	0.05
A2 <— Behavioral beliefs	0.82	0.05
A3 <— Behavioral beliefs	0.85	0.07
A4 <— Behavioral beliefs	0.85	0.06
A5 <— Behavioral beliefs	0.87	0.04
A6 <— Behavioral beliefs	0.64	0.06
A7 <— Behavioral beliefs	0.65	0.07
B1 <— Attitudes	0.94	0.02
B2 <— Attitudes	0.91	0.02
B3 <— Attitudes	0.91	0.03
B4 <— Attitudes	0.87	0.03
B5 <— Attitudes	0.86	0.03
C1 <— Normative beliefs	0.78	0.05
C2 <— Normative beliefs	0.81	0.05
C3 <— Normative beliefs	0.85	0.05
C4 <— Normative beliefs	0.85	0.06
C5 <— Normative beliefs	0.72	0.03
C6 <— Normative beliefs	0.60	0.06
D1 <— Subjective norms	0.91	0.02
D2 <— Subjective norms	0.92	0.03
D3 <— Subjective norms	0.85	0.04
D4 <— Subjective norms	0.85	0.04
E1 <— Control beliefs	0.77	0.06
E2 <— Control beliefs	0.87	0.06
E3 <— Control beliefs	0.83	0.08
E4 <— Control beliefs	0.83	0.06
E5 <— Control beliefs	0.70	0.10
E6 <— Control beliefs	0.64	0.12
E7 <— Control beliefs	0.28	0.14
F1 <— Behavioral intentions	0.85	0.04
F2 <— Behavioral intentions	0.84	0.03
F3 <— Behavioral intentions	0.92	0.03
F4 <— Behavioral intentions	0.70	0.06
G1 <— Perceived behavioral control	0.24	0.26
G2 <— Perceived behavioral control	0.86	0.05
G3 <— Perceived behavioral control	0.85	0.05
G4 <— Perceived behavioral control	0.73	0.06

Source: Compiled by this study.

(3) Multivariate Normality Test

Kline [36] proposed a standard to determine whether the variable conforms to the standard of univariate normal distribution. If the skewness value is less than 2 and the kurtosis value is less than 7 in absolute values, the benchmark of multivariate normality is reached. As seen from Table 4, the skewness values in this study were all below 2, and the kurtosis values were all within the absolute value of the numerical range of 7. Thus, they all met the standard.

Table 4. Summary of Skewness and Kurtosis Data of Observed Variables.

Item (Variable)	Skewness	C.R. Judgment Value	Kurtosis	C.R. Judgment Value
G4	−0.18	−1.12	−0.46	−1.41
G3	0.04	0.24	−0.43	−1.32
G2	0.23	1.44	−0.79	−2.43
F4	−0.77	−4.71	2.06	6.31
F3	−0.35	−2.14	0.59	1.82
F2	−0.70	−4.29	2.28	7.00
F1	−0.81	−5.00	1.67	5.11
E5	−0.84	−5.15	0.67	2.0
E4	−0.25	−1.52	−0.76	−2.33
E3	−0.37	−2.26	−0.41	−1.24
E2	−0.30	−1.87	−0.53	−1.62
D4	−0.40	−2.47	−0.12	−0.38
D3	−0.64	−3.94	0.59	1.81
D2	−0.60	−3.66	0.39	1.21
D1	−0.52	−3.21	0.40	1.23
C4	−0.34	−2.09	−0.61	−1.87
C3	−0.24	−1.44	−0.50	−1.53
C2	−0.34	−2.09	0.40	1.21
C1	−0.07	−0.44	−0.64	−1.98
B4	−0.53	−3.23	0.86	2.63
B3	−0.50	−3.08	0.36	1.09
B1	−0.57	−3.49	0.94	2.88
A5	−1.06	−6.49	2.10	6.46
A4	−0.75	−4.58	0.37	1.12
A3	−0.62	−3.81	−0.26	−0.79
A1	−0.88	−5.39	1.55	4.76
Multivariate			318.1	62.78

Source: Compiled by this study.

(4) Verification of Convergent Validity

Factor loading values are generally required to be greater than 0.70 in the normal criteria of convergent validity, and the factor loading values of most of the items in this study were greater than 0.70; thus, the convergent validity of the measurement model is good. Moreover, regarding the two indicators of the reliability of convergent validity, (1) composite reliability was >0.60 and (2) the average variance extracted was >0.50; thus, if the latent variable is >0.60, it means that the quality of the internal research model is good, and an average variance extracted >0.50 means that the quality of the internal latent variable is very good [33,34,37]. The compositional reliability values of each variable in this study were between 0.86 and 0.93, the values of the average variance extracted were between 0.66 and 0.81, and the Cronbach's α values for internal consistency fell between 0.75 and 0.92, which indicates that the intrinsic quality of the model has very good convergent validity, as shown in Table 5.

Table 5. Confirmatory Factor Analysis Summary of Theory of Planned Behavior Scale.

Model Parameter Estimates								Convergent Validity	
Latent Variable	Observed Variable	Unstandardized Factor Loading	Standard Deviation	C.R	Cronbach's α	Factor Loading	SMC	Compositional Reliability	Average Variance Extracted
Behavioral beliefs	A1	1.00			0.89	0.79	0.63	0.91	0.71
	A3	1.34	0.09	14.97	0.87	0.87	0.76		
	A4	1.28	0.09	14.69	0.87	0.87	0.75		
	A5	1.10	0.08	14.14	0.87	0.84	0.70		
Attitudes	B1	1.00			0.87	0.92	0.85	0.93	0.81
	B3	1.03	0.04	23.34	0.88	0.92	0.85		
	B4	0.89	0.05	19.35	0.92	0.86	0.74		
Normative beliefs	C1	1.00			0.87	0.79	0.63	0.90	0.69
	C2	1.10	0.08	13.83	0.86	0.83	0.69		
	C3	1.24	0.08	14.79	0.85	0.87	0.76		
	C4	1.26	0.09	13.87	0.87	0.84	0.70		
Subjective norms	D1	1.00			0.90	0.91	0.83	0.93	0.78
	D2	1.09	0.05	22.86	0.90	0.92	0.84		
	D3	1.03	0.05	18.78	0.91	0.85	0.72		
	D4	1.02	0.06	18.59	0.92	0.85	0.72		
Control beliefs	E2	1.00			0.83	0.85	0.72	0.89	0.66
	E3	1.07	0.07	15.65	0.83	0.84	0.71		
	E4	0.95	0.06	15.71	0.83	0.85	0.73		
	E5	0.82	0.07	11.85	0.88	0.70	0.49		
Behavioral intentions	F1	1.00			0.86	0.85	0.73	0.90	0.69
	F2	0.83	0.05	16.11	0.85	0.84	0.71		
	F3	0.96	0.05	18.50	0.82	0.92	0.85		
	F4	0.80	0.07	11.95	0.90	0.70	0.49		
Perceived behavioral control	G2	1.00			0.76	0.86	0.74	0.86	0.67
	G3	0.95	0.06	15.51	0.75	0.85	0.72		
	G4	0.78	0.06	12.18	0.85	0.73	0.54		

Source: Compiled by this study.

(5) Fit Analysis

The overall model fit was tested with reference to the eight indicators of [33]. Bagozzi and Yi believed that the smaller the χ^2 value, the better the ratio of χ^2 to degrees of freedom [38,39], and when the comparative fit index (CFI), the Tucker–Lewis index (TLI), and the standardized root mean square residual (SRMR) value are small, it means that they are within the tolerance range [40–43]. Browne and Cudeck stated that the closer the root mean square error of approximation (RMSEA) is to one, the better; Hair et al. stated the same for the goodness of fit index (GFI) and adjusted goodness of fit index (AGFI) values [40]. In this research structure model, the χ^2 value was 620.96; the ratio of χ^2 to degrees of freedom was 2.14, conforming to the standard of less than 3. CFI = 0.94, TLI = 0.93, SRMR = 0.07, RMSEA = 0.07, GFI = 0.84, and AGFI = 0.80; thus, the test results of the eight basic indicators in this study all passed the standard threshold of the SEM model fit [34], as shown in Table 6.

Table 6. Overall Model Fit Analysis.

Fit Indices	Allowable Range	This Research Model	Modified Bollen–Stine Bootstrap	Model Fit Judgment
χ^2 (Chi-square)	The smaller, the better	620.96	419.10	
Ratio of χ^2 to degrees of freedom	<3	2.14	1.45	Conforming
CFI	>0.90	0.94	0.98	Conforming
TLI	>0.90	0.93	0.97	Conforming
SRMR	<0.50	0.07	0.07	Conforming
RMSEA	<0.08	0.07	0.04	Conforming
GFI	>0.90	0.84	0.92	Conforming
AGFI	>0.90	0.80	0.90	Conforming

(6) Bollen–Stine Model Correction

Bollen [44] pointed out that in the structural equational modeling (SEM) analysis, if the number of study samples is more than 200, the chi-square value is likely to be too large,

resulting in a poor fit. Hence, the bootstrap analysis can be used to correct the fit index [45]. The number of samples in this study was 226, which might cause the fit index to fail to reach the ideal value. Therefore, this study used the Bollen–Stine *p*-value correction method built into AMOS 24.0 statistical software to perform *p*-value correction. The indicators after conducting the correction analysis of the model fit are shown in Table 6. The chi-square value was reduced from 620.96 of the estimated value of the maximum likelihood method to 419.10, and the ratio of chi-square to degrees of freedom was reduced from 2.14 to 1.45. All the fit indices met the test requirements.

3.4. Discussion

It can be seen from Figure 2 and Table 7 that H1 was supported, meaning the behavioral beliefs of pickleball players had a significant impact on attitudes. The results of this study are the same as the results of [28], and the possible reason is that pickleball players could have a sense of accomplishment or self-challenge, as well as other important results, from participating in the pickleball event; thus, they had a positive evaluation of participating in pickleball events. H2 was supported, meaning the normative beliefs of pickleball players had a significant impact on subjective norms. The results of this study are similar to the research conclusions of [29], and the possible reason is that when the participants of the pickleball event perceived that significant others or other pickleball lovers were supporting the behavior of participating in the event, these pickleball lovers would make suggestions or discuss the game strategy for the event. H3 was supported, meaning the control beliefs of pickleball players had a significant impact on perceived behavioral control. The results of this study are similar to the research results of [46], and the possible reason is that the higher the cognition of the pickleball players of their ability to compete and the number of opportunities or barriers, the stronger their ability to control their own competition strength. H4 was not supported, meaning the attitudes of pickleball players did not have a significant impact on behavioral intentions. This study result is different from the research results of [29], and the possible factor is that the pickleball players had positive attitudes toward participating in the event; however, their willingness to participate may be lowered due to other barriers. H5 was supported, meaning the subjective norms of pickleball players had a significant impact on behavioral intentions. This study result is the same as those of [30], and the possible reason is that the pickleball players' willingness to participate may be affected by recommendations from other people around them or other pickleball players (such as the competition level or grading system). H6 was supported, meaning the perceived behavioral control of pickleball players had a significant impact on behavioral intentions. This research conclusion is the same as that of [47]; that is, when the pickleball players had a high degree of mastery of physical fitness, technology, and tactics, they were more likely to participate in pickleball competitions, as shown in Table 7.

Table 7. Empirical Results of Research Hypotheses.

Hypotheses	Path Relationship	Path Value	Hypothesis Supported or Not
1	Behavioral beliefs -> Attitudes	0.87 *	Supported
2	Normative beliefs -> Subjective norms	0.79 *	Supported
3	Control beliefs -> Perceived behavioral control	0.83 *	Supported
4	Attitudes -> Behavioral intentions	0.14	Not supported
5	Subjective norms -> Behavioral intentions	0.55 *	Supported
6	Perceived behavioral control -> Behavioral intentions	0.31 *	Supported

* $p < 0.05$.

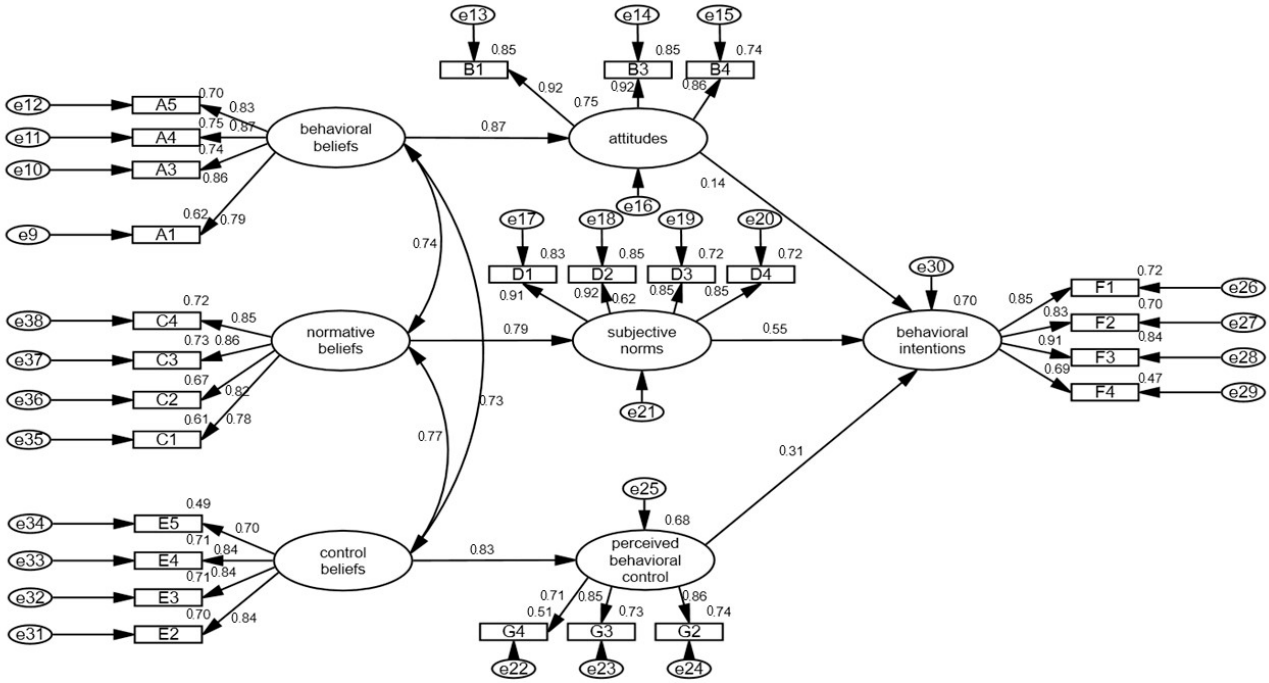

Figure 2. Diagram of Behavioral Intention Patterns of Pickleball players.

4. Conclusions and Suggestions

4.1. Conclusions

Based on the empirical analysis results, with the exception of H4, which is not supported, all other hypotheses are supported. The results of this study show that pickleball participants' normative beliefs had a significant impact on their subjective norms. From the perspective of sustainability, when holding future events, relevant units should evaluate the impact of the event on the local area. Integrating the concept of sustainability into the events will be helpful for the organizers to promote relevant events in the future, which can improve the entertainment value of the events and the image of the host city, promote cultural exchanges, and stimulate the development of the local economy, tourism, and accommodation industries. When there are more events to participate in, pickleball players can be more devoted to training, and more family members and friends will become familiar with the events, which will lead to more support from important others.

4.2. Suggestions

This study puts forward some suggestions for reference based on the above research results:

(1) For Players

The results of this article show that the control beliefs of pickleball players had a significant impact on their perceived behavioral control. Therefore, it is suggested that the participants must be self-demanding and constantly improve their physical fitness and skills in pickleball. Especially with the advancements in network technology today, besides practicing with other pickleball players, they can also use pickleball broadcasts or teaching videos on the Internet as materials to improve their own skills; for example, the PrimeTime Pickleball channel, which has many subscribers, provides diversified content, such as technical training, tactics, play, and physical fitness. The explanation and demonstration of professional pickleball players help to improve self-control beliefs, and then the accumulated strength can be used more completely in routine practices or official competitions. In addition, the results of this study also point out that the perceived behavioral control of pickleball players had a significant impact on their behavioral intentions. Thus, it is suggested that pickleball players can spend their spare time making friends with other

pickleball players in order to keep themselves closely connected to the sport; for example, joining a local pickleball club is a feasible way to engage in practice and pair exercises with club members to improve one's game, and some clubs have exchange games with other clubs from time to time. Through participation in such small activities, pickleball players will enjoy more opportunities for specialization and have a better grasp of the game when their abilities and skills are improved, thereby increasing their ambition to participate in pickleball events. In addition, the results of this study show that the behavioral beliefs of pickleball players had a significant impact on attitudes. It is suggested that pickleball players can further understand the benefits of pickleball in the process of participating in pickleball games, such as making themselves healthier, making more friends, and improving their own skills, in order to strengthen their positive attitude regarding participation in pickleball, and in turn, obtain physical challenges, pleasure, and satisfaction.

(2) For Event Organizers or Pickleball Promotion Units

The results of this study show that the subjective norms of pickleball players had a significant impact on behavioral intentions. Therefore, it is suggested that event organizers or pickleball promotion units invite domestic and foreign pickleball players or coaches to hold seminars or workshops from time to time. Through the sharing and transfer of professional knowledge, learning can be supported for pickleball players. Once enough models are provided, there will be more opportunities to transform this professional knowledge into real actions; for example, a virtual workshop has been established in which people can watch the course content of its training camp online. Moreover, it also provides dynamic activities for people to download from the Pickleball Activity Stations Guide. Through the abovementioned methods, more pickleball enthusiasts can gain more professional knowledge about pickleball from event organizers and pickleball promotion units, which will further enhance their willingness to play pickleball and their intention to participate in pickleball competitions. In addition, the results of this study point out that the normative beliefs of pickleball players had a significant impact on subjective norms. Therefore, it is suggested that event organizers or pickleball promotion units regularly hold related events, as such events will not only improve the cohesion of pickleball players but also allow the players to realize that there are people who support and make efforts to further increase participation in this sport and that the behavior of players participating in the event is endorsed; examples of such events are the National Championship Series (NCS), the USA Pickleball National Championships, and the USA Pickleball Diamond Amateur Championship. In other words, local competitions include events for both professional and amateur players, as well as regional competitions and national competitions. The diversity of competitions enhances the visibility of pickleball, which allows more players to participate in pickleball competitions, facilitates promotion by competition organizers, encourages pickleball players to devote themselves to training in response to different levels of competitions, and eventually leads to gaining the support of important others.

(3) Suggestions for Future Research

It can be seen from the research results that Hypothesis 4 of this study was not supported; that is, the attitudes of pickleball players had no significant impact on their behavioral intentions, and the standardized regression coefficient was low. It is suggested that in future research, researchers can use interviews or field observations to collect empirical data from the players of different competition levels in an extensive manner to further understand whether the level of competition, the location of the event, or even the current impact of COVID-19 has led to changes in the attitudes of the pickleball players toward competitions. The concept of recreation specialization, as developed by [48], can even be added to such exploration. Recreation specialization includes cognitive dimensions, such as environmental characteristics, skills, and knowledge; behavioral dimensions of past experiences of activities, familiarity with the environment, and the equipment owned; and emotional dimensions of pleasure and continued involvement in the activity. The analyses based on relevant theories can be conducive to the predictive analysis of the behavioral

intentions of pickleball players, and practical suggestions are proposed based on the results of empirical analysis.

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